MILITARY SPECIFICATION

BURS, BONE SURGERY, Wire Pass Drill

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wire pass drill style surgical burs for use with a medium bur guard in a pneumatic bone drill.

1.2 Classification. Wire pass drill, surgical burs shall be of the following sizes as specified (see 6.2).

Head Diameter

Size 1 - (1.5 mm)
Size 2 - (1.0 mm)
Size 3 - (2.0 mm)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

AMSC N/A FSC 6515

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medeical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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GG-I-526 Instruments, Dental and Surgical; General Specifications.
PPP-B-601 Boxes, Wood, Cleated-plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 Boxes, Shipping, Fiberboard.

Military

MIL-B-121 Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-116 Methods of Preservation.
MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151 Metal, Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.
MIL-STD-147 Palletized Unit Loads.
MIL-STD-794 Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM E18-Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E92-Standard Test Method for Vickers Hardness of Metallic Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documehnts also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The burs shall be in accordance with applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Material. The burs shall be fabricated from type 17-4 precipitation hardening corrosion-resisting steel.

3.3 Hardness. The burs shall be properly hardened and passivated. After proper stress relieving, the burs shall have a Rockwell hardness of C40 to C44, when tested as specified in 4.4.2.

3.4 Style, design and dimensions. The burs shall be of one piece construction. The burs shall be wire pass drill style compatible for use with medium bur guard component of PNEUMATIC BONE DRILL, National Stock Number (NSN) 6515-01-116-8197. The proximal end of the bur shall be round and non-notched. The blades and teeth of the burs shall be uniformly cut, shaped, and sharpened to cut at the tip as well as the sides of the head and provide automatic chip clearance. The burs shall conform to figure 1 for style, design and dimensions for the applicable NSN and size.

3.5 Runout. The total indicated runout of the burs shall not exceed .001 inch when measured as specified in 4.3.2.

3.6 Concentricity and straightness. The burs shall meet the concentricity and straightness specified in figure 1.

3.7 Autoclavability. The burs shall not show evidence of corrosion when tested as specified in 4.4.3. Boil test of GG-I-526 shall not apply.

3.8 Marking. The burs shall be marked with a safeline as specified in figure 1.

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3.9 Finish. The finish of the burs shall be as specified in figure 1. The operative end shall have a fine honed finish with sharp cutting edges. The safeline shall have an etched finish.

3.10 Workmanship. The burs shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

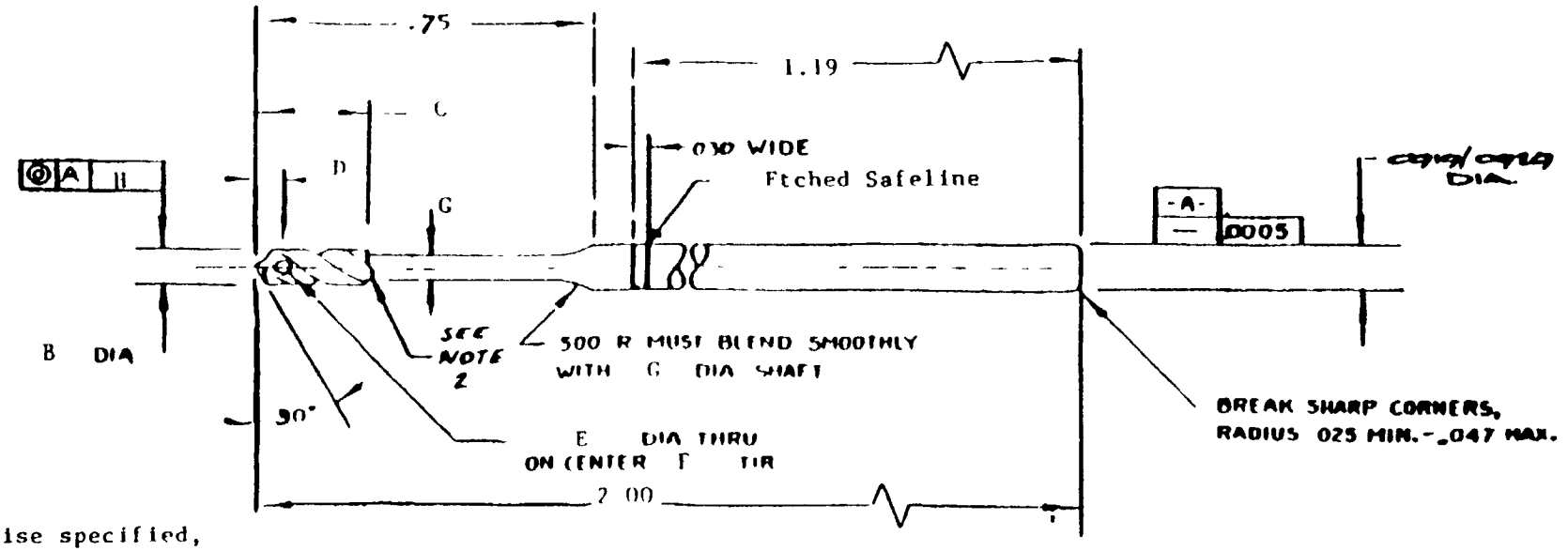
4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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Unless otherwise specified, tolerances are.

.XX ± .02
.XXX ± .010
Angles ± 0° 30'

	SIZE	B	C	D	E	F	G	H
6515-01-139-7741	1	.060 ± .005	.250	.063	.030 ±.005	.005	.051 ±.005	001
6515-01-139-7744	2	.040 +.002 -.000	.375 ± 060	.047	.020 .017	.002	.040 +.002 -.000	.002
6515-01-139-7745	3	.080 ± 005	.250	.063	.036 ± .005	.005	.063 ±.005	.001

NOTES.
1. Head Shape: RH Cutting, End Cutting, 2 teeth, 20° RH Spiral teeth to cut across center.
2. For Style 1. .12 Flute sweepout permissible on .051 diameter shank.
   For Style 3: .12 Flute sweepout permissible on .063 diameter shank
3. Finish #32
4. Remove all burrs
5. Unless otherwise specified, dimensions are in inches

FIGURE 1 Bur, bone surgery, wire pass drill

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4.1.4 Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of burs of the same size, manufactured under essentially the same conditions. Each unit package of burs shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

4.2 Quality conformance inspection. The examinations required to assure conformance of those burs or lots of burs to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3). The tests required for acceptance are specified in table II of 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as shown in table I. The unit of product for examination purposes shall be the individual bur.

TABLE I. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.5
Major B	S-1	2.5
Minor	II	4.1

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be as stated in the table.

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TABLE II. Sampling criteria for tests

Characteristics	Requirements	Test procedures	Inspection levels
Chemical Composition	3.2	4.4.1	see 4.2.3
Hardness	3.3	4.4.2	S-2
Autoclaving	3.7	4.4.3	S-1

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

4.3 Examination. The burs shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the following classification of defects listed in table IV. Examination shall not be restricted to the classified defects shown in table III.

TABLE III. Classification of defects

Category	Defect
Critical:	None defined.
Major A:	
101	Style and design not as specified.
102	Bur size or head shape not as specified.
103	Total indicated runout exceeds 0.001 inch.
104	Head diameter exceeds specification requirements at maximum cross section.
105	Etched safeline missing.
Major B:	
141	Operative end and shank dimensions not as specified (except for 104)
Minor:	
201	Cutoff end on shank not as shown in applicable figure.
202	Burs not free from defects such as chipped blades, cracks, or pits.
203	Finish on bur not as specified.
204	Overall length of bur not within specification limits.
205	Instructions for use inaccurate, inadequate, or missing.

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4.3.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

4.3.2 Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360 degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.3.3 Visual examination. Examination for possible bur defects not perceptible to the unaided eye shall be made using a 10X magnification.

4.4 Test. Test shall be conducted to determine compliance with specification requirements. Test procedures shall be as specified in GG-I-526 and as specified below:

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples, selected as specified in 4.2.3, shall be analyzed in accordance with Method 111 or 112 of Fed. Std. No. 151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

4.4.3 Autoclaving. The test burs shall be cleaned in accordance with the instructions supplied by the manufacturer. The burs shall then be autoclaved at 270 ± 5 degrees Fahrenheit (F.) and 27 ± 1 pounds per square inch pressure for a minimum of 20 minutes. After the burs are removed from the autoclave, the entire surface area of the bur shall be examined for evidence of corrosion.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

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5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Burs shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Burs shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative applications. Preservatives shall not be used.

5.1.1.4 Unit package. Each bur shall be packaged in a rigid plastic container having a telescopic friction or plug type closure. The bur shall be cushioned or immobolized within the container in such a manner as to prevent damage to the operative end.

5.1.1.5 Intermediate package. Twelve units containing burs of one size only shall be packaged in a snug fitting greaseproof, waterproof bag fabricated of MIL-B-121 barrier material. Closure of the bag shall be effected by heat sealing.

5.1.2 Level C. The level C preservation for burs shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages containing burs of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages containing burs of one size only, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for burs shall conform to MIL-STD-794 requirements for this level.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads commensurate with the level of packing specified in the contract or order shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.5.1 Levels A and B. Burs, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Burs, packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall include the size.

5.3.2 Levels A, B, and C. Each intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. Burs covered by this specification are intended for use in medical war readiness assemblages and DoD Deployable Medical Systems.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.
b. Size desired (see 1.2)
c. NSN (as applicable).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog.

National Stock Number	Item Idenfification
6515-01-139-7741	BUR, BONE SURGERY, Wire Pass Drill, 1.5 mm Head Diameter
6515-01-139-7744	BUR, BONE SURGERY, Wire Pass Drill, 1.0 mm Head Diameter
6515-01-139-7745	BUR, BONE SURGERY, Wire Pass Drill, 2.0 mm Head Diameter

MILITARY INTERESTS:

CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Review Activities:
Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITY
VA - OSS

PREPARING ACTIVITY:
DoD-MB

AGENT:
DLA-DM

Project No. 6515-3246P

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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